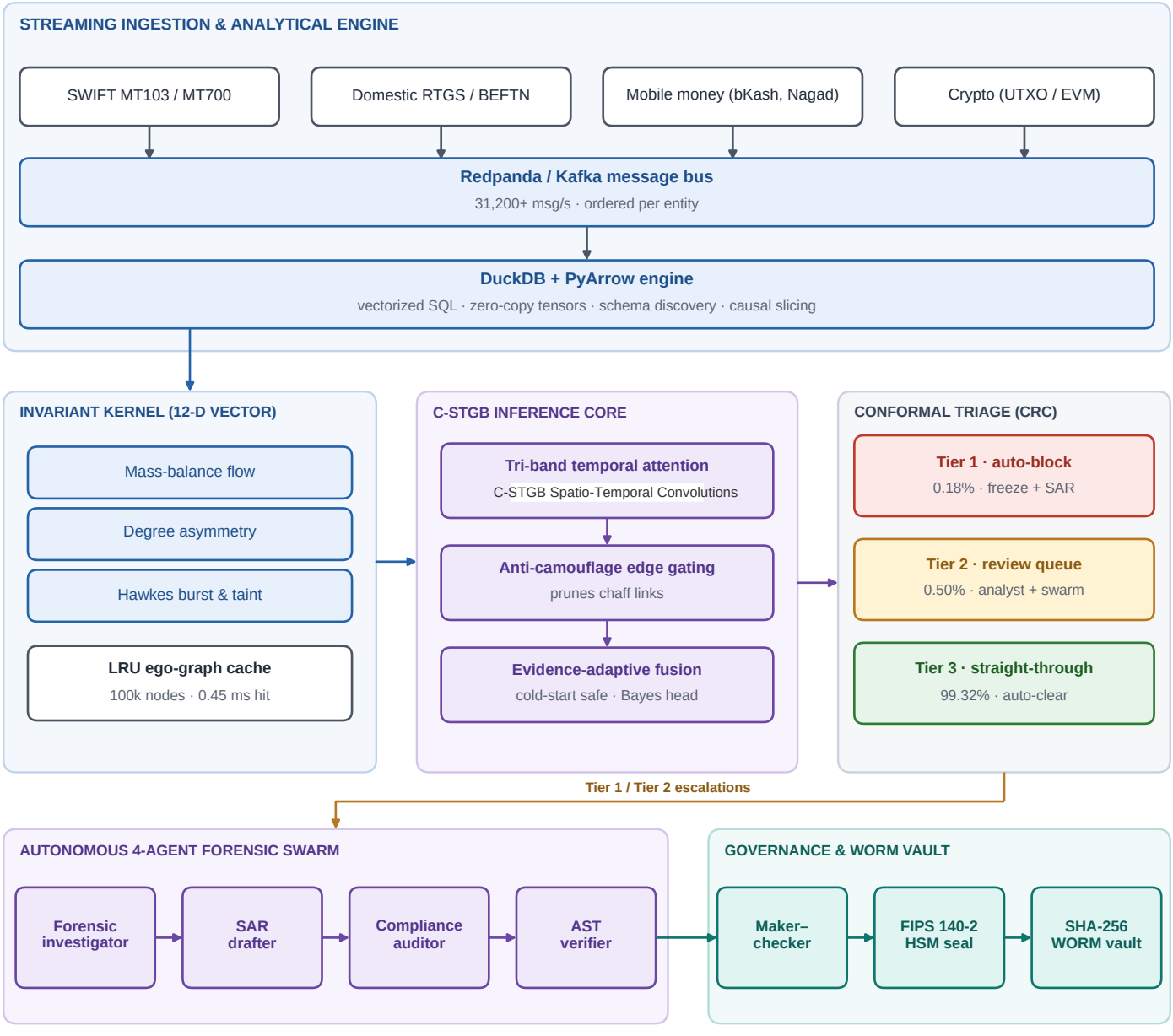
STREAMING INGESTION & ANALYTICAL ENGINE
SWIFT MT103 / MT700
Domestic RTGS / BEFTN
Mobile money (bKash, Nagad)
Crypto (UTXO / EVM)
Redpanda / Kafka message bus
31,200+ msg/s · ordered per entity
DuckDB + PyArrow engine
vectorized SQL · zero-copy tensors · schema discovery · causal slicing
INVARIANT KERNEL (12-D VECTOR)
Mass-balance flow
Degree asymmetry
Hawkes burst & taint
LRU ego-graph cache
100k nodes · 0.45 ms hit
C-STGB INFERENCE CORE
Tri-band temporal attention
C-STGB Spatio-Temporal Convolutions
Anti-camouflage edge gating
prunes chaff links
Evidence-adaptive fusion
cold-start safe · Bayes head
CONFORMAL TRIAGE (CRC)
Tier 1 · auto-block
0.18% · freeze + SAR
Tier 2 · review queue
0.50% · analyst + swarm
Tier 3 · straight-through
99.32% · auto-clear
Tier 1 / Tier 2 escalations
AUTONOMOUS 4-AGENT FORENSIC SWARM
Forensic investigator
SAR drafter
Compliance auditor
AST verifier
GOVERNANCE & WORM VAULT
Maker–checker
FIPS 140-2 HSM seal
SHA-256 WORM vault